

Srednicki QFT: Chapter 37

Canonical Quantization of Spinor Fields I
Cadabra2 expressions and numerical verification

Generated by `ch37_export_latex.py`

Contents

1	Canonical Anticommutation Relations	2
1.1	Equal-time CARs	2
1.2	Origin from Legendre transform	2
2	Mode Expansion of the Weyl Field	2
2.1	Plane-wave expansion	2
2.2	Operator CARs	3
3	Basis Spinors $u^s(p)$ and $v^s(p)$	3
3.1	Weyl equations	3
3.2	Explicit construction at rest	3
3.3	Massless case	3
3.4	Completeness relations	4
4	Deriving the CARs from the Mode Expansion	4
5	Normal-Ordered Hamiltonian	4
5.1	Classical Hamiltonian density	4
5.2	Mode expansion form	4
5.3	Normal ordering	5
6	Spin-Statistics Theorem	5
7	Summary	6

1 Canonical Anticommutation Relations

1.1 Equal-time CARs

The canonical anticommutation relations (CARs) for a left-handed Weyl field $\psi_\alpha(x)$ are (Srednicki eq. 37.3):

$$\boxed{\{\psi_\alpha(\mathbf{x}, t), \psi^{\dagger\dot{\beta}}(\mathbf{y}, t)\} = \sigma^0_\alpha{}^{\dot{\beta}} \delta^{(3)}(\mathbf{x} - \mathbf{y}) = \delta_\alpha{}^{\dot{\beta}} \delta^{(3)}(\mathbf{x} - \mathbf{y})} \quad (37.3)$$

where $\sigma^0 = I_2$ so the RHS is just $\delta_\alpha{}^{\dot{\beta}}$. The remaining equal-time anticommutators vanish:

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta(\mathbf{y}, t)\} = 0, \quad (37.4a)$$

$$\{\psi^{\dagger\dot{\alpha}}(\mathbf{x}, t), \psi^{\dagger\dot{\beta}}(\mathbf{y}, t)\} = 0. \quad (37.4b)$$

Cadabra2 expression for the LHS:

$$\psi_\alpha \psi^{\dagger\dot{\alpha}} + \psi^{\dagger\dot{\alpha}} \psi_\alpha = \delta_\alpha{}^{\dot{\alpha}} \delta^3(x - y)$$

1.2 Origin from Legendre transform

The kinetic Lagrangian $\mathcal{L}_{\text{kin}} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi$ gives the conjugate momentum:

$$\pi^\alpha = \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_\alpha)} = i\psi^{\dagger\dot{\beta}} \bar{\sigma}^0_{\dot{\beta}}{}^\alpha = i\psi^{\dagger\alpha} \quad (\text{since } \bar{\sigma}^0 = I_2) \quad (1)$$

Cadabra2: $\pi^\alpha = i_{\dot{\alpha}} \bar{\sigma}^{0\dot{\alpha}\alpha} = i\psi^{\dagger\alpha}$.

The canonical equal-time anticommutation $\{\psi_\alpha(\mathbf{x}), \pi^\beta(\mathbf{y})\} = i\delta_\alpha^\beta \delta^{(3)}(\mathbf{x} - \mathbf{y})$ then directly yields eq. (37.3).

✓ $\sigma^0 = I_2$: max error 0.0e + 00.

2 Mode Expansion of the Weyl Field

2.1 Plane-wave expansion

The left-handed Weyl field is expanded as (eq. 37.7):

$$\psi_\alpha(x) = \int \frac{d^3p}{(2\pi)^3} \sum_s \left[b_s(\mathbf{p}) u_\alpha^s(\mathbf{p}) e^{ip \cdot x} + d_s^\dagger(\mathbf{p}) v_\alpha^s(\mathbf{p}) e^{-ip \cdot x} \right] \quad (37.7)$$

with hermitian conjugate

$$\psi^{\dagger\dot{\alpha}}(x) = \int \frac{d^3p}{(2\pi)^3} \sum_s \left[b_s^\dagger(\mathbf{p}) \tilde{u}^{\dot{\alpha}s}(\mathbf{p}) e^{-ip \cdot x} + d_s(\mathbf{p}) \tilde{v}^{\dot{\alpha}s}(\mathbf{p}) e^{+ip \cdot x} \right]. \quad (2)$$

Here $p \cdot x = p^\mu x_\mu$ with mostly-plus metric; on shell $p^0 = \omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$.

2.2 Operator CARs

The creation/annihilation operators satisfy (eq. 37.8):

$$\{b_s(\mathbf{p}), b_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{ss'}, \quad (37.8a)$$

$$\{d_s(\mathbf{p}), d_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{ss'}, \quad (37.8b)$$

$$\{b_s, b_{s'}\} = \{d_s, d_{s'}\} = \{b_s, d_{s'}\} = \cdots = 0. \quad (37.8c)$$

3 Basis Spinors $u^s(p)$ and $v^s(p)$

3.1 Weyl equations

The basis spinors satisfy the Weyl equations (eq. 37.5):

$$p_{\alpha\dot{\alpha}} \tilde{u}^{\dot{\alpha}s}(p) = +m u_\alpha^s(p), \quad (37.5a)$$

$$p_{\alpha\dot{\alpha}} \tilde{v}^{\dot{\alpha}s}(p) = -m v_\alpha^s(p), \quad (37.5b)$$

where $p_{\alpha\dot{\alpha}} = p_\mu \sigma_{\alpha\dot{\alpha}}^\mu$ is the momentum matrix in spinor space.

3.2 Explicit construction at rest

At $\mathbf{p} = 0$, $p^\mu = (m, 0, 0, 0)$, the momentum matrix is:

$$p_{\alpha\dot{\alpha}} = p_\mu \sigma_{\alpha\dot{\alpha}}^\mu = -m \sigma_{\alpha\dot{\alpha}}^0 = -m I_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Basis spinors at rest (spin index $s = \pm$):

$$u_\alpha^+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{spin-up, particle})$$

$$u_\alpha^- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{spin-down, particle})$$

$$v_\alpha^+ = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{spin-up, antiparticle})$$

$$v_\alpha^- = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad (\text{spin-down, antiparticle})$$

3.3 Massless case

For a massless particle with $p^\mu = (E, 0, 0, E)$:

$$p_{\alpha\dot{\alpha}} = -E \sigma^0 + E \sigma^3 = \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix}$$

Eigenvalues: $0, -2$. The null eigenvector (helicity eigenstate u^+) lies in the kernel of $p_{\alpha\dot{\alpha}}$, consistent with the massless Weyl equation $p_{\alpha\dot{\alpha}} \tilde{u}^{\dot{\alpha}} = 0$.

3.4 Completeness relations

Summing over spins (eq. 37.10–37.11):

$$\sum_s u_\alpha^s(p) \tilde{u}^{\dot{\beta}s}(p) = -p_\alpha^{\dot{\beta}} + m \delta_\alpha^{\dot{\beta}}, \quad (37.10)$$

$$\sum_s v_\alpha^s(p) \tilde{v}^{\dot{\beta}s}(p) = -p_\alpha^{\dot{\beta}} - m \delta_\alpha^{\dot{\beta}}. \quad (37.11)$$

Numerical check at rest: $-p_{\alpha\dot{\alpha}} + m I_2 = 2m I_2$,

$$\sum_s u^s \otimes u^{s\dagger} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \approx \frac{1}{2m} \cdot \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \checkmark$$

Vector bilinears at rest:

$$\tilde{u}^{\dot{\alpha}+} \bar{\sigma}_{\dot{\alpha}\alpha}^\mu u_\alpha^+ = [1.0, 0.0, 0.0, -1.0], \quad \tilde{u}^{\dot{\alpha}-} \bar{\sigma}_{\dot{\alpha}\alpha}^\mu u_\alpha^- = [1.0, 0.0, 0.0, 1.0]$$

4 Deriving the CARs from the Mode Expansion

Inserting eq. (37.7) into $\{\psi_\alpha(x), \psi^{\dagger\dot{\beta}}(y)\}$ and using the operator CARs (37.8a–b) (eq. 37.12):

$$\begin{aligned} \{\psi_\alpha(x), \psi^{\dagger\dot{\beta}}(y)\} &= \int \frac{d^3p}{(2\pi)^3} \sum_s u_\alpha^s(\mathbf{p}) \tilde{u}^{\dot{\beta}s}(\mathbf{p}) e^{ip(x-y)} + \int \frac{d^3p}{(2\pi)^3} \sum_s v_\alpha^s(\mathbf{p}) \tilde{v}^{\dot{\beta}s}(\mathbf{p}) e^{-ip(x-y)} \\ &= \int \frac{d^3p}{(2\pi)^3} \left[(-p_\alpha^{\dot{\beta}} + m \delta_\alpha^{\dot{\beta}}) e^{ip(x-y)} + (-p_\alpha^{\dot{\beta}} - m \delta_\alpha^{\dot{\beta}}) e^{-ip(x-y)} \right]. \end{aligned} \quad (3)$$

The momentum-dependent terms cancel between the two integrals, leaving:

$$\{\psi_\alpha(x), \psi^{\dagger\dot{\beta}}(y)\} \Big|_{x^0=y^0} = \delta_\alpha^{\dot{\beta}} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad \checkmark \quad (4)$$

5 Normal-Ordered Hamiltonian

5.1 Classical Hamiltonian density

From the Weyl Lagrangian via Legendre transform:

$$H = \int d^3x \left[-i\psi^\dagger \bar{\sigma}^k \partial_k \psi + \frac{1}{2} m \psi \psi + \frac{1}{2} m \psi^\dagger \psi^\dagger \right]. \quad (5)$$

5.2 Mode expansion form

Inserting the mode expansion (eq. 37.17):

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} \left[b_s^\dagger(\mathbf{p}) b_s(\mathbf{p}) - d_s(\mathbf{p}) d_s^\dagger(\mathbf{p}) \right]. \quad (6)$$

5.3 Normal ordering

Using $\{d_s(\mathbf{p}), d_s^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^{(3)} \delta_{ss'}$:

$$-d_s d_s^\dagger = +d_s^\dagger d_s - \{d_s, d_s^\dagger\} = +d_s^\dagger d_s - (2\pi)^3 \delta^{(3)}(0). \quad (7)$$

Discarding the (divergent, negative) zero-point energy:

$$\boxed{: H := \int \frac{d^3 p}{(2\pi)^3} \omega_{\mathbf{p}} \left[b_s^\dagger(\mathbf{p}) b_s(\mathbf{p}) + d_s^\dagger(\mathbf{p}) d_s(\mathbf{p}) \right]} \quad (37.17)$$

Cadabra2: $: H := \int \frac{d^3 p}{(2\pi)^3} (\omega_p b_s^\dagger(p) b_s(p) + \omega_p d_s^\dagger(p) d_s(p))$

Key contrast with bosons:

Field type	Zero-point energy	$: H :$
Boson (scalar)	$+\frac{\omega}{2}$ per mode	$\int \omega a^\dagger a$
Fermion (Weyl)	$-\frac{\omega}{2}$ per mode	$\int \omega (b^\dagger b + d^\dagger d)$

6 Spin-Statistics Theorem

The spin-statistics theorem (Pauli 1940) states:

$$\text{Spin-}\frac{1}{2} \implies \text{anticommuting quantization.} \quad (8)$$

Proof sketch via microcausality: for a Lorentz-invariant theory,

$$[\mathcal{O}(x), \mathcal{O}'(y)] = 0 \quad \text{for spacelike } (x - y)^2 > 0.$$

If a spin- $\frac{1}{2}$ field were quantized with *commutators*:

$$[\psi_\alpha(x), \psi^{\dagger\dot{\beta}}(y)] = \int \frac{d^3 p}{(2\pi)^3} [(-p_\alpha^{\dot{\beta}} + m) e^{ip(x-y)} - (-p_\alpha^{\dot{\beta}} - m) e^{-ip(x-y)}] \neq 0 \quad (\text{spacelike})$$

— a violation. With anticommutators the p -terms cancel and the result vanishes outside the light cone, as required.

7 Summary

Object	Expression
Equal-time CARs	$\{\psi_\alpha(\mathbf{x}, t), \psi^{\dagger\dot{\beta}}(\mathbf{y}, t)\} = \delta_\alpha^{\dot{\beta}} \delta^{(3)}(\mathbf{x} - \mathbf{y})$
Conjugate momentum	$\pi^\alpha = i\psi^{\dagger\alpha}$
Mode expansion	$\psi_\alpha = \int \frac{d^3p}{(2\pi)^3} \sum_s [b_s u_\alpha^s e^{ipx} + d_s^\dagger v_\alpha^s e^{-ipx}]$
Operator CARs	$\{b_s, b_{s'}^\dagger\} = \{d_s, d_{s'}^\dagger\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{ss'}$
Weyl eq. (massive)	$p_{\alpha\dot{\alpha}} \tilde{u}^{\dot{\alpha}s} = +m u_\alpha^s, \quad p_{\alpha\dot{\alpha}} \tilde{v}^{\dot{\alpha}s} = -m v_\alpha^s$
Completeness	$\sum_s u_\alpha^s \tilde{u}^{\dot{\beta}s} = -p_\alpha^{\dot{\beta}} + m \delta_\alpha^{\dot{\beta}}$
Normal-ordered H	$:H := \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} [b^\dagger b + d^\dagger d]$
Spin-statistics	Spin- $\frac{1}{2}$ requires anticommuting fields (microcausality)

Next: Chapter 38 develops the spinor algebra toolkit (σ -completeness, trace identities, Fierz rearrangement) needed for explicit Feynman diagram calculations.